Surgical Endoscopy (2024) 38:6193–6202
https://doi.org/10.1007/s00464-024-11131-z

NEW TECHNOLOGY

Evaluation of forces applied to tissues during robotic-assisted surgical tasks using a novel force feedback technology

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Received: 20 April 2024 / Accepted: 27 July 2024 / Published online: 12 September 2024
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Abstract

Background The absence of force feedback (FFB) is considered a technical limitation in robotic-assisted surgery (RAS). This pre-clinical study aims to evaluate the forces applied to tissues using a novel integrated FFB technology, which allows surgeons to sense forces exerted at the instrument tips.

Methods Twenty-eight surgeons with varying experience levels employed FFB instruments to perform three robotic-assisted surgical tasks, including retraction, dissection, and suturing, on inanimate or ex-vivo models, while the instrument sensors recorded and conveyed the applied forces to the surgeon hand controllers of the robotic system. Generalized Estimating Equations (GEE) models were used to analyze the mean and maximal forces applied during each task with the FFB sensor at the "Off" setting compared to the "High" sensitivity setting for retraction and to the "Low", "Medium", and "High" sensitivity settings for dissection and suturing. Sub-analysis was also performed on surgeon experience levels.

Results The use of FFB at any of the sensitivity settings resulted in a significant reduction in both the mean and maximal forces exerted on tissue during all three robotic-assisted surgical tasks ($p < 0.0001$). The maximal force exerted, potentially associated with tissue damage, was decreased by 36%, 41%, and 55% with the use of FFB at the "High" sensitivity setting while performing retraction, dissection, and interrupted suturing tasks, respectively. Further, the use of FFB resulted in substantial reductions in force variance during the performance of all three types of tasks. In general, reductions in mean and maximal forces were observed among surgeons at all experience levels. The degree of force reduction depends on the sensitivity setting selected and the types of surgical tasks evaluated.

Conclusions Our findings demonstrate that the utilization of FFB technology integrated in the robotic surgical system significantly reduced the forces exerted on tissue during the performance of surgical tasks at all surgeon experience levels. The reduction in the force applied and a consistency of force application achieved with FFB use, could result in decreases in tissue trauma and blood loss, potentially leading to better clinical outcomes in patients undergoing RAS. Future studies will be important to determine the impact of FFB instruments in a live clinical environment.

Keywords Force feedback · Haptic technology · Robotic-assisted surgery · Inanimate model · Ex-vivo tissue

Robotic-assisted surgery (RAS) represents a revolution in minimally invasive surgery (MIS). In the last twenty years, RAS has been adopted in a wide variety of procedures in various surgical specialties. RAS essentially offers advantages similar to conventional MIS (traditional endoscopic/laparoscopic surgery) while improving vision, precision, dexterity, and ergonomics through technological advances [1–4]. Some of the clinical benefits of RAS over laparoscopic and open surgery include a shorter length of hospital stay, less blood loss, and faster recovery [2, 4–10]. A commonly recognized limitation associated with minimally invasive surgery including RAS is the absence of haptic feedback. Even after reaching proficiency in the learning curve, there is a possibility that the lack of haptic feedback may result in excessive force exerted, leading to unintentional

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tissue damage during surgery, which could have a negative clinical impact on patients, including more pain and prolonged recovery time [11–13]. As a way of compensating for the lack of haptic feedback while performing RAS, surgeons train themselves to employ visual cues to determine the forces they are applying on tissue [14, 15], a fact that has been recognized as a major contributing factor to the learning curve in the performance of RAS [16–18].

Haptic feedback generally describes touch feedback and includes both kinesthetic (force) and cutaneous (tactile) feedback [11]. Haptics pertains to force, distributed pressure, temperature, vibrations, and texture, some of which are difficult to quantify, acquire, and display [11, 19–21]. The concept of "true haptic perception" describes a sense of feedback that does not entail a change in the somatosensory modality of the perception. In contrast, the opposite, pseudo-haptic feedback, such as vibrotactile or visual haptics, forces the recipient to integrate a different somatosensory modality for perception [22, 23]. Determining the type of haptic feedback that causes minimal disruption to the modality of perception is one of the challenges facing the development of haptic feedback technology. In human–machine interactions such as those encountered in the performance of RAS, coupling the sensor feedback to the same mode of feedback naturally used by humans would be ideal. As we experience push-and-pull forces when interacting with everyday objects, sensing these forces might be compatible with the feedback we naturally experience in our daily lives [24].

In the past decade, scientists and engineers have dedicated substantial effort to the creation of haptic feedback instruments used in RAS and laparoscopic surgery [16, 22, 23, 25, 26]. However, only a limited number of robotic-assisted surgical systems with integrated haptic technology are currently available. One such robotic surgical system with integrated haptic technology is indicated for use in total knee and hip replacement surgeries [27]. Used as an adjunct to pre-surgical planning, the feedback can be triggered and sensed by surgeons when they deviate from the planned surgery. When the surgeon attempts to move outside the pre-defined spatial boundary, the system provides mechanical resistance to prevent the surgeon from doing so [27]. While the application of this haptic technology has been impactful in orthopedic surgery showing positive results, such as reduced deviation from the pre-operative plan [27], there is more complexity involved in the integration of haptic technology in soft-tissue surgery [28]. The only other study examining the effect of FFB on RAS is described in a recent publication on a pneumatically driven robotic system (Saroa surgical system by Riverfield Inc., Tokyo, Japan), which shows that the use of a force feedback function can decrease the maximal force exerted during suturing tasks using a synthetic polyester experimental model [29].

In this study, we describe a novel integrated FFB technology in a commercially available robotic system, which employs instruments with built-in force sensors that allow surgeons to sense at the hand controllers the force applied at the instrument tips during RAS. This is a pre-clinical study that measured and analyzed the forces applied to the tissue using novel FFB instruments to perform common surgical tasks on inanimate models and ex-vivo animal tissues. We hypothesized that the utilization of FFB could reduce the forces exerted on tissue and enhance consistent force application during surgical tasks performed by surgeons at varying levels of RAS experience.

Materials and methods

System, instruments, and data recording

A compatible da Vinci Surgical System (Intuitive Inc., Sunnyvale, California, USA) and its accessories, such as cannulas, seals, and sterile drapes, along with instruments with built-in force sensors, were used in this study. These newly developed FFB instruments are similar to standard RAS instruments in form and function, except for the addition of force sensors at the instrument tips that measure push or pull forces (along the X, Y, and Z axes) and transmit these forces to the hand controllers at the surgeon console allowing the integration of force sensing with the use of the surgical system (Fig. 1). The FFB instruments do not provide grip feedback or tactile sensation. The feedback relayed to the surgeon's hand controls is scaled to three selectable sensitivity settings, namely Low, Medium, or High with Medium set as the default sensitivity setting.

The Force Feedback instruments used in this study included Fenestrated Bipolar Forceps, Maryland Bipolar Forceps, Large Needle Driver, Mega SutureCut Needle Driver, and Cadiere Forceps. The magnitude of the force applied to tissues was measured and recorded by the FFB instruments regardless of whether force feedback was disabled ("Off" setting), or which sensitivity setting was chosen at the surgeon's hand controllers to perceive the force feedback. The force recorded at the "Off" setting served as the baseline control, which was compared to the force exerted when instruments were used at different (Low, Medium, or High) FFB sensitivity settings. Force was measured in units of Newtons (N).

Study participants

Surgeons ($n = 28$) at various experience levels, namely novice ($n = 6$), intermediate ($n = 6$), and experienced ($n = 16$) from different surgical specialties participated in the study.

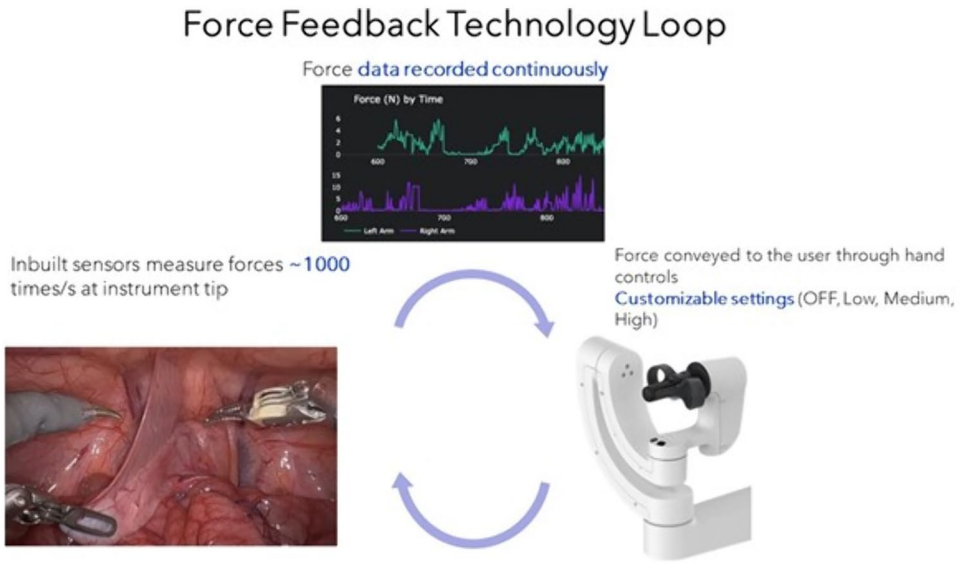


Fig. 1 Instruments with integrated Force Feedback Sensing technology. The force data are continuously recorded and conveyed to the surgeon user through hand controls

Fig. 2 Views from the surgeon console while performing retraction task using porcine tissue, dissection task using the Kotobuki model, and suturing task using porcine tissue

Surgeon experience levels were defined as follows: novice (had performed less than 50 RAS procedures), intermediate (had performed between 51 and 200 RAS procedures), and experienced (had performed more than 200 RAS procedures). The participating surgeons ($n = 28$) included surgeons specializing in urologic surgery ($n = 6$), general/colorectal surgery ($n = 16$), and thoracic surgery ($n = 6$). All surgeons consented to the participation in the study, signed a non-disclosure agreement and video disclosure form, and received financial compensation for their participation.

Study design

Initial surgeon training

The participating surgeons were provided with an overview of the study protocol followed by training on using the FFB Instruments. The training included practicing surgical tasks using these instruments on a plant-based surgical simulating organ marketed by KOTOBUKI Medical Inc., Saitama, Japan (https://kotobukimedical.com/news_en/2021/04/09/vtt-rbc-type-a-new-vtt-model-that-visualizes-thermal-diffusion/).

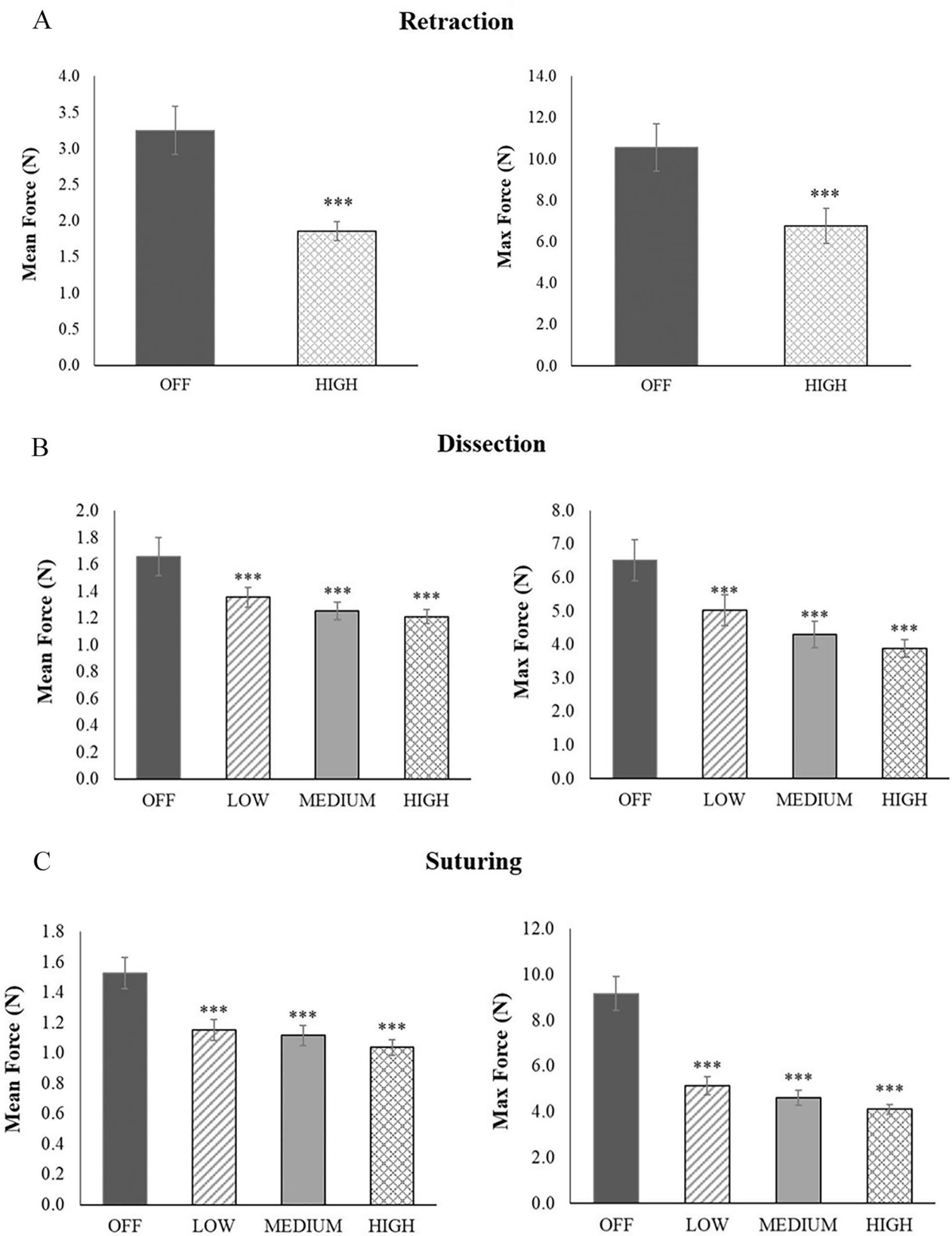
Surgical tasks, instruments, and models used

Surgeons performed three primary surgical tasks, namely dissection, retraction, and suturing (Fig. 2).

For the retraction task, participants created a peritoneal flap in the body wall that connected a pre-defined 2 × 2-inch square in ex-vivo porcine abdominal wall or human cadaveric tissue. The task was performed once in two different sensitivity settings (Off, High) using either the FFB Cadiere Forceps or FFB Fenestrated Bipolar Forceps.

For dissection tasks, participants dissected one simulated nodule embedded in the inanimate plant-based simulating organ (Kotobuki model) two times at each sensitivity setting for perceiving the force feedback (Off, Low, Medium, High). The instruments used to perform dissection were the FFB Maryland Bipolar Forceps and the FFB Fenestrated Bipolar Forceps.

For the interrupted suturing task, participants sutured two limbs of ex-vivo porcine small bowel in a side-to-side fashion using 3–0 silk sutures using the Lembert technique. A subset of surgeons used human cadavers to perform interrupted suturing tasks. The same sequence was completed a second time on another loop of the bowel. One stitch was defined as one surgeon knot followed by three additional half hitches. The task involved completing two stitches at each of


A
Retraction
Mean Force (N)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
***
OFF
HIGH
Max Force (N)
14.0
12.0
10.0
8.0
6.0
4.0
2.0
0.0
***
OFF
HIGH
B
Dissection
Mean Force (N)
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
***
***
***
OFF
LOW
MEDIUM
HIGH
Max Force (N)
8.0
7.0
6.0
5.0
4.0
3.0
2.0
1.0
0.0
***
***
***
OFF
LOW
MEDIUM
HIGH
C
Suturing
Mean Force (N)
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
***
***
***
OFF
LOW
MEDIUM
HIGH
Max Force (N)
12.0
10.0
8.0
6.0
4.0
2.0
0.0
***
***
***
OFF
LOW
MEDIUM
HIGH

◂**Fig. 3** Mean Force and Max Force applied to tissues during the performance of three surgical tasks at various FFB sensitivity settings, compared to the "Off" setting. **A** Comparison of the mean and maximal forces during retraction tasks ($n=28$) using porcine tissues. **B** Comparison of the mean and maximal forces during dissection tasks ($n=27$, one measurement was excluded as an outlier) using a Kotobuki model. **C** Comparison of the mean and maximal forces during interrupted suturing ($n=28$) using porcine tissue or cadaver models. *$p<0.05$, **$p<0.01$, and ***$p<0.001$

the Force Feedback sensitivity settings (Off, Low, Medium, High), which were repeated two times. The instruments used for suturing were the FFB Large Needle Driver or the FFB Mega SutureCut Needle Driver.

Statistical analyses

The FFB instruments recorded the magnitude of the force exerted on tissues during retraction, dissection, and suturing tasks. The average (mean) force and maximal (max) force were analyzed separately. Forces less than 0.5N were excluded from the analysis, as 0.5N has been reported to be the typical minimum force recorded during endoscopic surgeries [30]. There was one instance (a mean force measurement during dissection) where the recorded force was larger than two standard deviations above the mean force, which was considered an outlier and was excluded from the analysis.

Generalized Estimating Equations (GEE) models were used to analyze per-participant paired differences in mean and max force between the FFB "Off" setting and FFB 'On' at each individual sensitivity setting, i.e., Low, Medium, or High. The dependent variable was the per-participant paired differences from "Off". Two models were fitted for both absolute and percent differences from "Off". Least squared means, standard errors, and p-values were provided for each model. An F-test was utilized to assess differences in the variability of mean and max measured force between the FFB "Off" and Force Feedback "On" at Low, Medium, or High sensitivity settings to compare the respective variances. Kruskal–Wallis tests by ranks were used to compare the change in forces from "Off" to each sensitivity setting level (i.e., Low, Medium, or High) between surgeons at different experience levels. A p-value of less than 0.05 was considered statistically significant. All analyses were performed using the SAS software (version 9.4 of the SAS system).

Results

This pre-clinical study assessed the changes in the force applied to tissues at the different sensitivity settings used for FFB perception when compared to the force exerted at the "Off" setting (without FFB). A sub-analysis included an analysis of the force data based on surgeon experience levels to evaluate whether the impact of FFB on force reduction varied between surgeon experience levels.

Force feedback reduces both the mean and maximal forces applied

Significant reductions were observed in both the mean and max forces applied to tissues when the FFB was enabled compared to the corresponding no-feedback ('Off') conditions during the performance of all three types of surgical tasks (Fig. 3). During the performance of the retraction tasks ($n=28$) on the ex-vivo porcine abdominal wall, there was a significant reduction in the mean forces from 3.25 ± 1.77 N at the "Off" setting to 1.86 ± 0.70 N at the "High" setting ($p<0.0001$). In the case of the max force recorded by the FFB instrument, there was approximately a 36% decrease with FFB (10.57 ± 5.96 N at the "Off" setting vs 6.77 ± 4.41 N at the "High" setting) (Fig. 3A).

Similar reductions were observed when dissection ($n=27$) and suturing tasks ($n=28$) were performed. As shown in Fig. 3B, the mean force recorded during the dissection of the inanimate plant-based model was significantly lower with the use of FFB. The mean forces were reduced from 1.68 ± 0.75 N with FFB "Off" to 1.36 ± 0.38 N at the "Low" setting ($p<0.001$), 1.25 ± 0.34 N at the "Medium" setting ($p<0.0001$), and 1.21 ± 0.27 N at the "High" setting ($p<0.0001$). Compared to the maximal force exerted

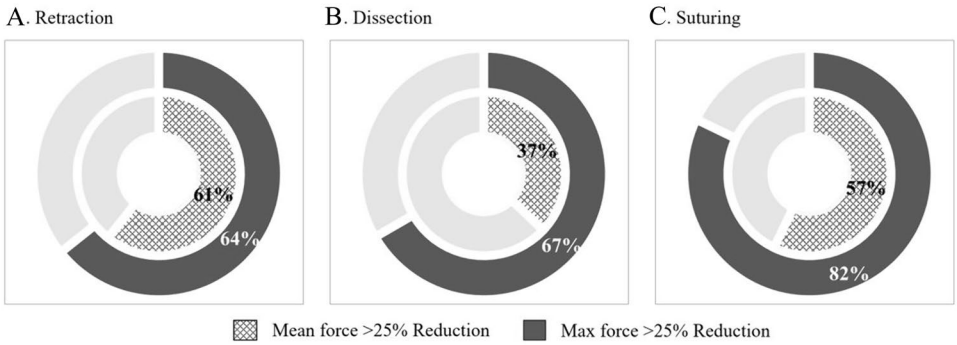


Fig. 4 Percentage of surgeons who experienced > 25% reduction in Mean and Max forces with FFB at the "High" setting compared to the "Off" setting. The analyses include **A** retraction tasks ($n=28$), **B** dissection tasks ($n=27$), and **C** suturing tasks ($n=28$)

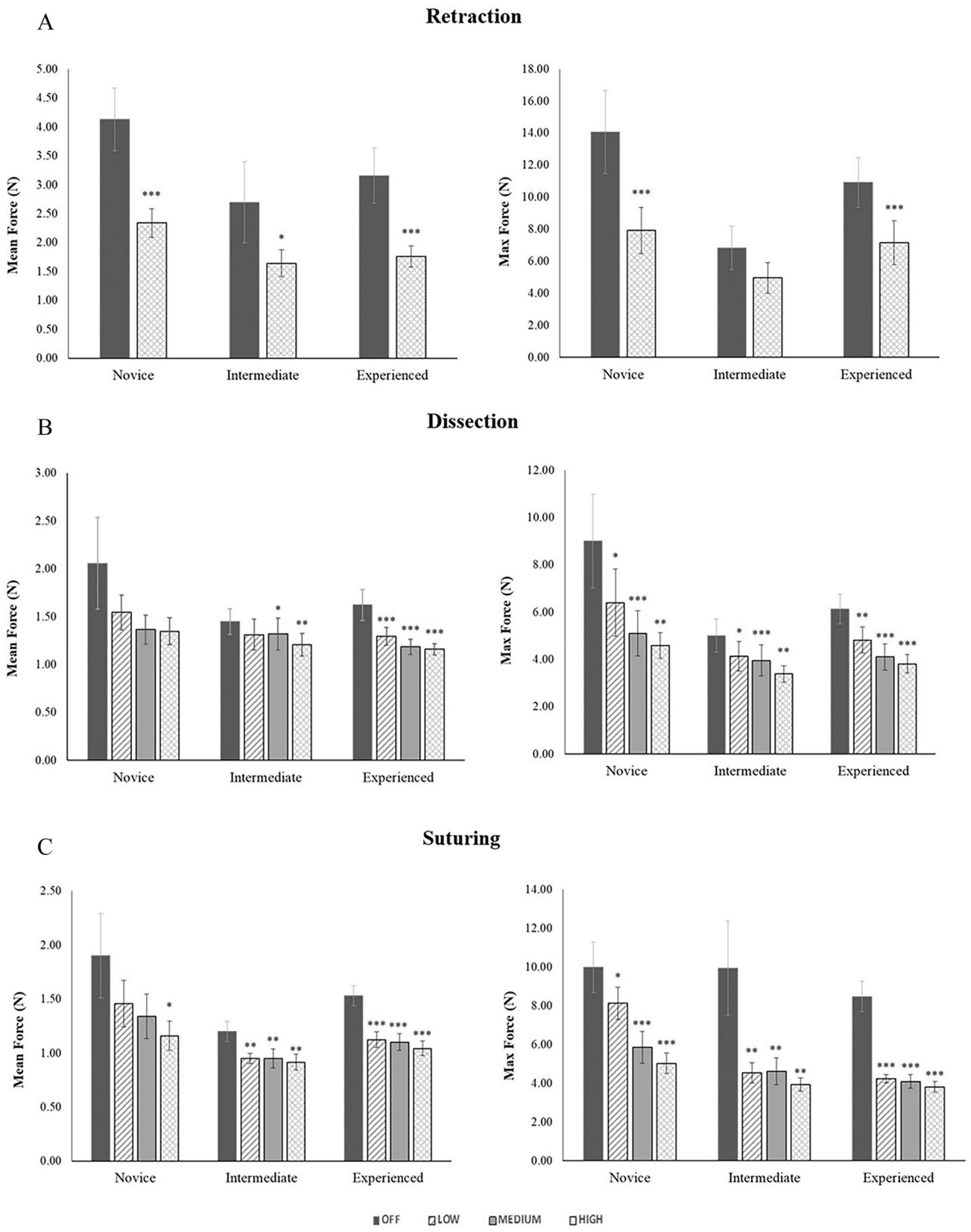


Fig. 5 Mean Force and Max Force applied by surgeon cohorts at different experience levels (novice $n=6$, intermediate $n=7$, experienced $n=15$) during retraction (**A**), dissection (**B**), and suturing (**C**) at various sensitivity settings, compared to those at the "Off" setting. $*p<0.05$, $**p<0.01$, and $***p<0.001$

Table 1 Mean Force reduction comparison using Kruskal–Wallis rank sum tests

Mean force reduction (N)	Surgeon experience level			P value (Kruskal–Wallis rank sum test)		
	Novice	Intermediate	Expert	N vs. I	N vs. E	I vs. E
Retraction	$n=6$	$n=7$	$n=15$			
OFF vs. HIGH, mean (SD)	1.8 (1.1)	1.1 (1.3)	1.4 (1.5)	0.199	0.243	0.751
Dissection	$n=6$	$n=6$	$n=15$			
OFF vs. HIGH, mean (SD)	0.7 (1.1)	0.2 (0.2)	0.5 (0.4)	0.150	0.755	**0.043**
OFF vs. MEDIUM, mean (SD)	0.7 (1.0)	0.0 (0.2)	0.4 (0.4)	**0.037**	0.815	**0.029**
OFF vs. LOW, mean (SD)	0.5 (1.0)	0.0 (0.2)	0.3 (0.3)	0.055	0.392	**0.016**
Suturing	$n=6$	$n=7$	$n=15$			
OFF vs. HIGH, mean (SD)	0.7 (0.8)	0.3 (0.3)	0.5 (0.4)	0.199	0.586	0.169
OFF vs. MEDIUM, mean (SD)	0.6 (1.0)	0.2 (0.3)	0.4 (0.4)	1.000	0.533	0.307
OFF vs. LOW, mean (SD)	0.4 (0.9)	0.2 (0.2)	0.4 (0.2)	0.668	0.755	0.130

Bold values indicate that have met the threshold for statistical significance: p<0.05

Table 2 Max Force reduction comparison using Kruskal–Wallis rank sum tests

Max force reduction (N)	Surgeon experience level			*P* value (Kruskal–Wallis rank sum test)		
	Novice	Intermediate	Expert	N vs. I	N vs. E	I vs. E
Retraction	$n=6$	$n=7$	$n=15$			
OFF vs. HIGH, mean (SD)	6.1 (4.4)	1.9 (2.4)	3.8 (3.7)	**0.046**	0.276	0.169
Dissection	$n=6$	$n=6$	$n=15$			
OFF vs. HIGH, mean (SD)	4.4 (3.9)	1.6 (1.5)	2.3 (1.9)	0.337	0.312	0.586
OFF vs. MEDIUM, mean (SD)	3.9 (2.7)	1.0 (0.7)	2.0 (1.8)	**0.037**	0.161	0.213
OFF vs. LOW, mean (SD)	2.6 (2.9)	0.9 (0.9)	1.3 (1.6)	0.150	0.276	0.697
Suturing	$n=6$	$n=7$	$n=15$			
OFF vs. HIGH, mean (SD)	4.9 (2.6)	6.0 (5.7)	4.7 (2.9)	0.886	0.586	0.972
OFF vs. MEDIUM, mean (SD)	4.1 (2.6)	5.3 (5.3)	4.4 (3.3)	0.775	0.876	0.916
OFF vs. LOW, mean (SD)	1.9 (2.0)	5.4 (5.8)	4.2 (2.8)	0.317	0.062	0.805

Bold values indicate that have met the threshold for statistical significance: p<0.05

at the FFB "Off" setting (6.60 ± 3.21N), significant reductions were seen in the maximal force exerted at the "Low" (5.02 ± 2.42N; 24% reduction; $p < 0.0001$), "Medium" (4.29 ± 2.04N; 35% reduction; $p < 0.0001$) and "High" (3.89 ± 1.39N; 41% reduction; $p < 0.0001$) settings in the performance of suturing (Fig. 3B).

During interrupted suturing tasks performed on ex-vivo porcine tissues, significant reductions were observed in both mean and maximal forces (Fig. 3C). Compared to the mean force of 1.53 ± 0.55 N without FFB, when FFB was enabled at the "Low," "Medium," and "High" sensitivity settings, the mean forces decreased to 1.15N ± 0.36N, 1.12 ± 0.35 N, and 1.04 ± 0.27 N, respectively. Additionally, using FFB at the Low, Medium, and High sensitivity settings resulted in significant reductions in the maximal force applied on tissue (by 44%, 50%, and 55%, respectively; $p < 0.0001$; Fig. 3C) compared to the maximal force recorded at the "Off" setting (9.16 ± 3.90 N; Fig. 3C).

Distribution of participants by force data reduction

Figure 4 illustrates the percentages of surgeons who experienced a reduction in mean and maximal forces of greater than 25% when FFB was sensed at the "High" setting compared to the "Off" setting. During retraction, 61% of surgeons showed a > 25% reduction in mean force, and 64% showed a > 25% reduction in maximal force. During dissection, 37% of surgeons showed a > 25% reduction in mean force, and 67% showed a > 25% reduction in maximal force exerted. In the case of suturing, 57% of surgeons showed a > 25% reduction in mean force, and 82% showed a > 25% reduction in maximal force exerted.

Assessment of force reduction based on surgeon's experience level

The data were further analyzed to assess if the effect of using FFB on force reduction varied among surgeons at different experience levels (Fig. 5). The three cohorts analyzed included novice ($n = 6$), intermediate ($n = 7$), and experienced ($n = 15$) surgeons. In the novice group, the mean force was significantly reduced during retraction and suturing tasks using FFB at the "High" setting ($p < 0.001$ and $p < 0.05$, respectively). The maximal force exerted by the novice group was also significantly reduced during dissection and suturing tasks. The degree of force reduction varied depending on the sensitivity setting used. In the intermediate experience group, there were significant reductions in the mean force during retraction and in both mean and maximal forces during dissection and suturing tasks compared to the "Off" setting without FFB. The data on the experienced group showed that using FFB at any of the three sensitivity settings resulted in a significant decrease in both mean and max forces consistently during all surgical tasks ($p < 0.001$).

In addition, Kruskal–Wallis rank sum tests were performed to compare the force reduction achieved at different experience levels, including novice vs. intermediate (N vs. I), novice vs. experienced (N vs. E), and intermediate vs. experienced (I vs. E) comparisons as shown in Table 1 (reduction in Mean Force) and Table 2 (reduction in Max Force). The only pairs of comparison where significant differences were observed included novice vs. intermediate and intermediate *vs.* experienced, with novice and Intermediate cohorts achieving significantly more reductions in mean force exerted during the performance of dissection $p < 0.05$; Table 1) and the novice cohort achieving more reductions in the maximal forces exerted compared to the intermediate cohort during the performance of retraction and dissection tasks ($p < 0.05$; Table 2).

Use of force feedback reduces the variance in forces applied

The use of FFB resulted in substantial reductions in force variance during all three types of surgical tasks. For the retraction task, there were significant reductions in force variance for the experienced cohort ($p < 0.01$) and intermediate ($p < 0.05$) cohort, while the reduction in force variance in the novice cohort was not statistically significant ($p = 0.11$). During the dissection task, there were significant reductions in force variance for experienced surgeons at the "Low" ($p < 0.05$), "Medium" ($p < 0.01$), and "High" ($p < 0.001$) sensitivity settings. At the same time, there were only significant reductions seen for novice surgeons when FFB was used at the "Medium" ($p < 0.05$) and "High" ($p < 0.05$) settings, while no significant differences were observed in the case of surgeons at the intermediate experience level. During suturing, a significant reduction in force variance was observed only in novice surgeons using FFB at the "High" sensitivity setting ($p < 0.05$).

Discussion

The absence of haptic feedback is the most commonly noted disadvantage associated with minimally invasive surgery including RAS, requiring the surgeons to rely on visual information to evaluate the forces being applied to tissue during RAS [2, 31]. Even after surgeons reach proficiency in performing RAS, the lack of a direct perception of the forces applied could increase the possibility of tissue damage [22]. A recent meta-analysis performed by Bergholz et al. which included 56 primary studies conducted from 2013 to 2023, compared RAS performed with and without any type of haptic feedback to quantify the benefits of restoring the haptic feedback [32, 33]. The results of this meta-analysis confirmed that haptic feedback in RAS was effective in reducing forces exerted and the completion time, as well as improving accuracy and success rates in the performance of surgical tasks [32, 33]. These findings from pre-clinical studies provide further evidence of the positive impact of haptic technology on surgical performance.

Our study evaluates the use of force feedback technology integrated into a commercially available new generation robotic surgical system on the forces applied on tissues during the performance of routine surgical tasks using inanimate or ex-vivo tissue models. The integrated force feedback technology allows surgeons operating at the surgeon console to feel at their hand controllers the push-and-pull forces exerted on the tissue with the use of instruments installed with force sensors. We hypothesized that using the FFB feature enables surgeons to avoid exerting excessive force or tension and thereby minimize the risk of tissue damage. In this pre-clinical study, using novel RAS surgical instruments with FFB sensors, surgeons at all three experience levels (novice, intermediate, and experienced) applied significantly less force (both mean force and max force) on tissue when performing common surgical tasks such as retraction, dissection, and suturing. The variance in force reduction was also significantly reduced when the FFB feature was utilized during all three surgical tasks. These findings are consistent with existing literature demonstrating reduced tissue damage, improved surgical technique, and fewer errors when performing simulated tasks with the use of force feedback [16, 22, 24–26]. This study is the first to measure the impact of FFB sensing integrated into a commercially available RAS system. The findings demonstrate reductions in the mean and maximal forces applied during the performance of common surgical tasks. Recently reported observations

on the reduction of maximal forces using FFB are from a study using the Saroa surgical system [29].

Importantly, our data show that in the performance of retraction, using FFB significantly reduces mean forces, max forces, and force variance compared to forces recorded without FFB. These findings, particularly the decrease in the maximal force applied during tissue retraction, suggest that using FFB can enable a "gentler surgery" and reduce potential tissue damage. In the performance of dissection tasks on the Kotobuki plant model, reductions were seen in mean and max force when the subjects were combined and across experience levels. Reducing the amount of force during the dissection has the potential to enhance safely performing surgery in proximity to critical organs and structures such as blood vessels. The data on suturing show similar force reductions with FFB. Ultimately, these benefits could translate to improved clinical outcomes in patients such as less tissue trauma, less bleeding, and quicker recovery time.

The analysis of force data based on surgeon experience showed that significant differences in the mean force reduction were seen in pairwise comparisons of intermediate *vs.* novice cohorts and intermediate *vs.* experienced surgeon cohorts during the performance of dissection tasks. Previous studies have indicated that novice surgeons experience greater reductions in forces compared to surgeons at higher experience levels [32, 33]. However, the findings of our study suggest that surgeons at all experience levels benefit from the use of the integrated FFB technology, as demonstrated by comparable reductions in forces achieved by novice, intermediate, and experienced surgeon cohorts. A potential concern for consideration is that the use of FFB technology might have a negative impact on experienced surgeons who have already mastered the application of visual haptics in the performance of RAS from years of experience. The results of this study, however, show that the expert surgeon group also achieved significant reductions in the mean and maximal forces applied and force variability with FFB, suggesting that the type of haptic feedback provided by the FFB feature augments their visuo-motor mapping skills.

This study had some limitations. The surgical tasks being evaluated were performed on pre-clinical models, which may not reflect the same conditions encountered while working with living tissues. Further, the sample size of participating surgeons was modest. Nonetheless, the findings showed statistically significant force reductions consistently across different surgical tasks in inanimate and ex-vivo tissue models. Further investigations with robust sample sizes in a real-world clinical setting are needed to better understand the impact of the FFB on the performance of RAS.

In conclusion, our study demonstrates that the utilization of FFB technology integrated into a commercially available next-generation robotic surgical system significantly decreases the forces applied in the performance of common surgical tasks by novice, intermediate, and experienced surgeons. This innovative technology has the potential to enable safer and gentler surgeries, resulting in better surgical outcomes in patients undergoing RAS.

Acknowledgements We would like to thank the following engineers and developers at Intuitive for their expertise in creating the FFB instruments: Aruna Krishnan, Yan Zhong, Katie Ewing, Xuechen Li, Shreya Purohit, Andrew Penfold, Todd Tourand

Funding Funded by Intuitive

Declarations

Disclosures Dr. Awad has received educational grants from Applied Medical, Baxter, Bard/BD, Ethicon, Intuitive, Medtronic, Styker and Storz. Drs. Blatnik and Schumaker report consultancy and speaker fees from Intuitive. Mika Padmanabhan-Kabana is a former employee of Intuitive. Dr. Raynor has no conflicts of interest to disclose.

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